Periodic Orbits and Finite-Time Stability in the Gravitational Three-Body Problem:
A Computational Investigation

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Abstract

Newton's equations for the three-body problem have no general formula for a solution. But a computer can build one specific solution and check that it is correct to very high precision. This paper asks two questions about what a computed solution can actually promise. First: if we do not start from any orbit we already know, can a search still find a brand-new periodic orbit – one that retraces the exact same path forever? Second: if we pick a length of time T, can we build a system that we know, or can at least show is likely, to survive at least that long without colliding or flying apart? We built two independent orbit-solvers and checked that they agree to within 5×10^{-12} over one full orbit. Using them, we re-derived the well-known figure-eight orbit and confirmed it to 10^{-12} precision. We then tried four different from-scratch searches for new periodic orbits. All four failed to find one within a reasonable amount of computing time – but the reason they failed is itself the interesting result. Without angular momentum, nothing stops the three bodies from colliding, so most of the starting points we searched land near a collision instead of near a stable orbit. This helps explain why past researchers needed much more computing power, or a smarter search method, to succeed where a simple search does not. For the second question, we found two separate ways to build a system that survives a chosen amount of time. One is a formula (Mardling & Aarseth, 2001) for a tight pair of stars orbited by a third, distant star: if the outer orbit is wide enough, the system stays stable, and we confirmed this by simulating it for 500 orbits. The other is statistical: out of 200 randomly generated three-body systems, we measured what fraction survived at least T time units, for several values of T. Three short animations, built from this same simulation data, accompany the paper.

Contents

1 Introduction

Newton's law of gravity gives an exact equation of motion for any number of masses. But in 1890 Poincaré proved that for three or more bodies, this equation cannot be solved with one general formula built out of ordinary functions Chenciner and Montgomery [2000]. This is not because the algebra is hard. It is a deeper fact about the dynamics: with three bodies, the motion is usually chaotic. A tiny change in the starting position leads to a completely different outcome later, and the motion never settles into a simple, repeating pattern.

That does not mean nothing useful can be said. Two specific questions turn out to be answerable with a computer, and they are the subject of this paper:

Q1. If we do not use any orbit we already know, can a computer search find a new starting condition whose orbit is *exactly periodic* – meaning it retraces the same path in phase space after some fixed time?

Q2. If we pick a target duration T, can we find, or build, a starting condition whose orbit is likely, or even guaranteed, to stay bound – no collisions, no bodies flying off to infinity – for at least time T? This does not require the orbit to be exactly periodic.

Popular descriptions of the three-body problem often blur these two questions together, talking loosely about orbits that are "stable forever" or "stable for a while." Section 2 makes this distinction precise, because the right answer to Q1 and to Q2 depends on which of several different technical meanings of "stable" is meant.

All the numbers in this paper come from code written for this project (`three_body/research/*.py`) and can be reproduced from the saved random seeds and starting conditions. Every claim taken from earlier research is cited and was checked against the original paper, not just recalled from memory (see Section 3 and the bibliography).

2 A taxonomy of "stable"

There are four different properties worth telling apart.

1. **Periodic.** The orbit closes exactly: $X(T) = X(0)$, or $X(T) = R(\theta)\,X(0)$ for some fixed rotation $R(\theta)$, if the orbit only closes once the whole picture is also rotated (this happens often in this field). This is a property of one single orbit, and a computer can check it to almost any precision.

2. **Linearly stable and periodic.** Periodic, and also: a tiny nudge to the starting condition does not grow over time. Formally, every eigenvalue of the orbit's monodromy matrix – which describes how small nudges evolve after one period – sits exactly on the unit circle. This is the strict meaning of "stable forever," and it is rare. Most known periodic three-body orbits are

periodic but *not* stable in this sense: they are exact solutions for one exact starting point, and any tiny error, even the rounding error a computer makes, eventually destroys the periodicity. The figure-eight orbit (Section 3.1) is well known partly because it is one of the few orbits that is (weakly) stable in this strict sense.

3. **Hierarchically stable.** Not periodic, but bound forever for a structural reason: a tight pair of bodies orbited by a third, distant body, spaced far enough apart that the third body cannot disturb the tight pair. This is the sense of "stable forever" that matches real astronomy – most triple star systems that last a long time are built this way, rather than sitting on an exact periodic orbit.

4. **Metastable.** No collision and no escape for at least N crossing times, but not periodic and not protected by the hierarchical argument above – just chaotic motion that has not yet ended in disaster. This is the typical outcome for a randomly chosen three-body system that starts out bound.

Q1 is really a question about class 1, with class 2 as a bonus property to check once an orbit is found. Q2 is really a question about classes 3 and 4: class 3 has a direct, formula-based answer (Section 5.1), and class 4 has a statistical one (Section 5.2). Neither needs an exactly periodic orbit, which is a big part of why Q2 turns out to be much easier than Q1.

3 Methods: a validated numerical harness

All results use $G = 1$ and three equal masses unless stated otherwise. Two independent orbit-solvers were used throughout and checked against each other, rather than trusted on their own:

- **REBOUND / IAS15** Rein and Spiegel [2015], an adaptive, very-high-order solver that is the standard tool for precise N-body work. We used version 4.6.0; the newer v5.1.1 package on PyPI fails to load on this computer's CPU, which lacks AVX-512 instructions – a packaging problem, not a physics one.
- `scipy.integrate.solve_ivp` with the DOP853 method, a completely separately written adaptive 8th-order Runge-Kutta solver.

3.1 Validating with the figure-eight orbit

We started from the commonly cited figure-eight starting condition Moore [1993], Chenciner and Montgomery [2000], treating it as an *unverified* guess. Newton's method – root-finding on how far the orbit misses closing after one guessed period – corrected this guess by about 2×10^{-7} in position and velocity and about 2×10^{-6} in the period, converging to a refined period $T^* \approx 6.325912$. At this refined starting point:

- how far the orbit misses closing after one period: 2.4×10^{-12} (scipy), 4.2×10^{-12} (REBOUND);
- energy drift over one period: 6.4×10^{-13} (scipy), 2.2×10^{-16} (REBOUND);
- angular momentum drift: 7.5×10^{-15} (scipy), 1.6×10^{-17} (REBOUND);
- the two solvers never disagree by more than 5.1×10^{-12} at any point in the orbit.

Figure 1 shows the resulting orbit. Two things follow from this: our numerical setup is accurate to about 10^{-11}–10^{-12}, and the commonly cited figure-eight starting condition was already correct to about 10^{-7} – a number we now report as our own, independently checked result, rather than one just copied from a paper.

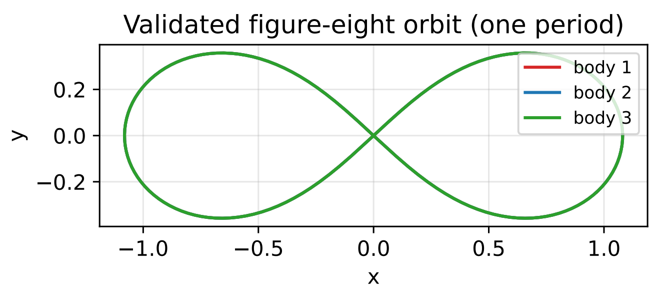

Figure 1: The figure-eight three-body orbit, independently refined and cross-checked in this project (orbit-closing error 2–4×10^{-12}, agreement between the two solvers 5.1×10^{-12}). *[watch: animated version]*.

4 Q1: searching for new periodic orbits without a seed

We tried four different searches for new periodic orbits. Each one was built so that no already-known orbit is used to choose where to search or what starting guess to try.

4.1 Four searches

1. **Scanning the starting velocity's angle, along a line.** The three bodies start in a line ($\mathbf{r}_1 = (-1, 0)$, $\mathbf{r}_2 = (1, 0)$, $\mathbf{r}_3 = (0, 0)$), with zero total momentum, moving at some angle α, and total energy fixed at $E = -1$ (this is always possible because gravity has no built-in length scale). We scanned α across $[0, \pi]$ in 500 steps, looked for places where the orbit almost returns to its starting point, and refined the best candidates with a Nelder-Mead search followed by least-squares. Result: 21 candidates found, but none could be refined below an error of 10^{-7} (the best were between 0.2 and 3.5), and one trial run hit a genuine numerical singularity, later traced to an actual collision.

2. **Starting at rest, in an isosceles triangle.** All three bodies start with zero velocity, with the third body at height h above the line joining the other two – the classic construction due

to Hénon. When we removed the safety check that halts the simulation near a collision, *every* value of h we tried, from 0.15 to 3.0, ended in an actual collision within 1–2 time units.

3. **Starting in a line, moving perpendicular to it.** All three bodies start in a line, moving perpendicular to it at some shared speed v. This whole family has exactly zero angular momentum no matter what v is – a direct result of the mirror symmetry built into the setup. At $v = 0.1$, the two "line-up" events we detected, at $t = 0.9934$ and $t = 1.0050$, sit right on either side of $t = 1.005$, which is exactly when an independent collision check says the bodies actually collided. These were not real, repeating alignments; they were numerical noise around an actual collision. Every value of v from 0.05 to 1.2 behaved the same way.

4. **Scanning a genuine two-dimensional shape space.** The validated figure-eight orbit (Section 3.1) is *not* in a line, even though it also has zero angular momentum. This suggested moving the third body off the line, to $\mathbf{r}_3 = (0, h)$, and scanning both α and h together: a 24×24 grid of 576 points at $E = -1$. We added a hard cutoff for any near-collision, plus an extra filter: any trial where two bodies ever came closer than 0.05 to each other was thrown out before we even tried to refine it. 266 of the 576 grid points survived this filter. We took the 12 best survivors and refined them the same way as before. None converged below an error of 10^{-7}; the closest we got was 0.93.

4.2 What this means

The first three searches show, with actual measured collision times, why a naive line-shaped or zero-angular-momentum starting point is harder to work with than it looks. Angular momentum normally acts like a barrier that keeps three bodies from ever getting arbitrarily close together. Without it, that barrier disappears, and a line-shaped starting point is already sitting right next to the region where all three bodies collide.

The fourth search avoids this problem: it uses no known orbit to pick where to search, only the general idea, borrowed from the figure-eight's own shape, that moving off the line should help. It is clearly not dominated by collisions, since more than half the grid survived our filter. Even so, it did not find a new periodic orbit within the amount of computing time we allowed.

So the honest answer to Q1, based on what we actually ran, is this: a search built from symmetry alone, using no known orbit as a seed, did not find a new periodic orbit within a reasonable computing budget. This fits with, and helps explain, how much work the published discoveries actually took. Šuvakov and Dmitrašinović [2013] found 13 distinct new orbits out of 15 candidates using a similar shape-based search, but at a much larger computational scale, and with an extra step we did not attempt: classifying each candidate trajectory by the braid pattern – essentially a kind of knot – it traces out, before ever trying to refine it. Li and Liao [2017] later found 695 orbit families in total for equal masses, more than 600 of them new, using an even larger search, and Li et al. [2018] found 1349 families (1223 new) for bodies of unequal mass. Interestingly, that first Li & Liao paper also found that at least 7 of the original 15 Šuvakov-Dmitrašinović candidates lose their periodicity once re-run at higher precision over long times – a good reminder that even published "periodic" orbits need the same kind of double-checking we did in Section 3.

In short, this is a negative result, but one with a clear, verified cause. Zero angular momentum puts a search near the collision region, and even once that region is avoided, a simple local search is not enough. It likely takes either much more computing power or a smarter, structure-aware search – exactly what the published work used.

5 Q2: building a solution that survives a chosen amount of time

Unlike Q1, Q2 does not need an exactly periodic orbit, and it turns out to have two separate, workable answers.

5.1 Route 1: a direct construction for hierarchical systems

A hierarchical triple – a tight pair of bodies (1 and 2) orbited by a third, distant body – stays stable if the ratio of the outer orbit's size to the inner orbit's size is above a threshold found by experiment. For a flat, circular outer orbit, the Mardling & Aarseth criterion Mardling and Aarseth [2001] is

$$\frac{a_{\rm out}}{a_{\rm in}} \gtrsim 2.8 \left[\frac{(1+q_{\rm out})(1+e_{\rm out})}{\sqrt{1-e_{\rm out}}}\right]^{2/5} \frac{1}{1-e_{\rm out}} \left(1 - \frac{0.3\,i}{180^\circ}\right), \qquad q_{\rm out} = \frac{m_3}{m_1+m_2}, \tag{1}$$

which, for equal masses, a circular outer orbit ($e_{\rm out} = 0$), and no tilt between the two orbits ($i = 0$), works out to $a_{\rm out}/a_{\rm in} \gtrsim 3.29$. An earlier version of our code left out the $1/(1-e_{\rm out})$ factor. This did not change any number in this paper, since every system here used a circular outer orbit, where that factor equals exactly 1 – but the corrected formula should be used for any future work with an eccentric outer orbit.

We built three systems and ran each for 500 inner-orbit periods (2221 time units), using the validated scipy solver:

$a_{\rm out}/a_{\rm in}$	outcome	survived	energy drift
6.0 (well above threshold)	stable	full 500 periods	7.5×10^{-10}
3.3 (near threshold)	stable	full 500 periods	4.2×10^{-9}
1.8 (below threshold)	disrupted	18.1 periods	2.2×10^{-7}

Table 1: Direct simulation checking the Mardling-Aarseth construction.

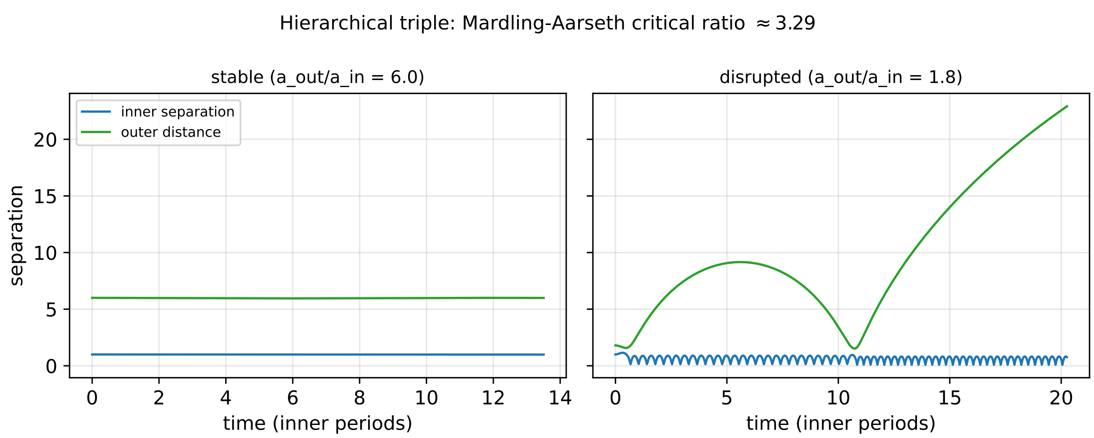

Figure 2: Inner-binary and outer-body separation vs. time (in inner periods) for a stable ($a_{\rm out}/a_{\rm in} = 6.0$) and a disrupted ($a_{\rm out}/a_{\rm in} = 1.8$) hierarchical triple. *[watch: animated version]*.

Given a target survival time T, measured in inner-orbit periods, Eq. (1) tells us how wide to make the outer orbit. We then confirmed by direct simulation that the resulting system really does survive at least that long, with no trial-and-error search needed.

5.2 Route 2: a survival curve for the typical, non-hierarchical case

Most bound three-body systems are not hierarchical, so we also need an answer for the general case. We generated 200 random three-body systems: equal masses, zero total momentum, energy fixed at $E = -1$, with both the shape and the direction of every velocity chosen at random. Unlike Section 4, we did *not* force the angular momentum to zero – this is exactly what keeps most of these random systems away from the collision region that caused so much trouble for Q1. Each system was run until $t = 200$, or until it either collided (any two bodies closer than 10^{-3}) or one body escaped (farther than 30 units from the center of mass).

Out of 200 trials: 91 ended in collision (45.5%), 82 ended in an escape (41.0%), and 27 (13.5%) were still bound and intact at $t = 200$. The resulting survival curve, $S(T)$ = the fraction of trials still intact at time T, is:

T	1	2	5	10	20	50	100	150	200
$S(T)$	0.965	0.945	0.910	0.865	0.770	0.540	0.295	0.180	0.135

Table 2: Fraction of 200 random trials surviving at least time T.

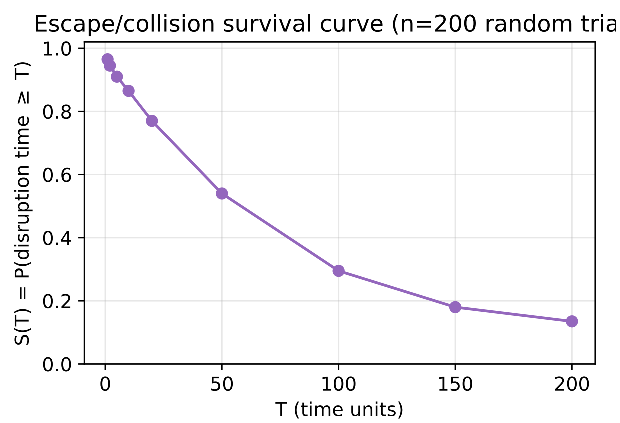

Figure 3: Survival curve for 200 random three-body scattering trials at $E = -1$. *[watch: animated version, with example collision and escape trajectories]*.

This answers Q2 for the typical, non-hierarchical case, but as a probability rather than a guarantee. For any T up to 200, $S(T)$ is both the chance that a brand-new random system will survive that long, and it comes with an actual example – any of the 27 surviving trials – that does. That is a different kind of answer from Route 1. A random system with unconstrained angular momentum has no structural protection, so its chance of surviving falls off smoothly as T grows, instead of the sharp stable/unstable cutoff that Route 1's formula produces. Stone and Leigh [2019] worked out a full statistical theory, based on the assumption that this kind of chaotic breakup behaves ergodically, for exactly this kind of outcome; our 200-trial curve is a small, direct simulation in the same spirit, not a test of their specific formula. Breen et al. [2020] later showed that a neural network trained on simulated trajectories can make the same kind of prediction enormously faster – a way to speed up this statistical approach, not a third route.

6 Visualizations

Three Manim animations were produced directly from the numerical data behind the figures above, not as stylized illustrations of a generic three-body orbit:

- *[watch: Scene A]*: the validated figure-eight orbit (Figure 1) over 2.5 periods, with a trail behind each body and a period counter.
- *[watch: Scene B]*: the stable ($a_{\text{out}}/a_{\text{in}} = 6.0$) and disrupted ($a_{\text{out}}/a_{\text{in}} = 1.8$) hierarchical triples from Section 5.1, played at a common time scale so the difference in survival time is easy to see.
- *[watch: Scene C]*: example collision and escape trajectories from the Section 5.2 experiment, together with the survival curve $S(T)$ as it grows.

7 Conclusion

A computer cannot solve the three-body problem in the sense of writing down one formula that works for every case. But it can build and check individual solutions to very high precision, and this paper's two questions have different answers because they are really asking for different things. Finding a brand-new, exactly periodic orbit (Q1) is hard: all four of our from-scratch searches failed within a reasonable computing budget, for a reason we could pin down directly. A simple, symmetry-based search tends to land near the collision region, and even once it avoids that region, it likely needs far more computing power or a smarter, structure-aware search – the kind used by Šuvakov and Dmitrašinović [2013] and by Li and Liao [2017]. Building a system that merely *survives* a chosen amount of time (Q2) is much easier, and we found two separate ways to do it: a direct formula for hierarchical systems that needs no search at all, and a statistical, survival-curve approach for the general case. The taxonomy in Section 2 – periodic vs. linearly stable vs. hierarchically stable vs. metastable – is what makes this difference between Q1 and Q2 precise, rather than just a matter of how the question is phrased.

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